

Experiment 4

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1 Title : Study of γ -ray energy spectrum using a scintillation counter with Multi Channel Analyzer (SCA).

2 Aim :

- Finding channel numbers corresponding to photopeaks of Co^{60} 's decay to find relation between channel number and energy.
- Finding energy of γ -ray in the decay of Cs^{137} .
- Finding photopeak efficiency of Co^{60} .
- Studying variation of resolution for two Co^{60} peaks with variation of PMT voltage.

3 Source : We took a radioactive isotope of cobalt ($_{27}Co^{60}$) & cesium ($_{55}Cs^{137}$) as our γ -ray source.

4 Theory :

4.1 Gamma (γ) decay :

4.1.1 Cobalt-60: It has a half life of 5.27 years. Undergoes β decay in mainly 2 steps releasing 2 different γ -rays :

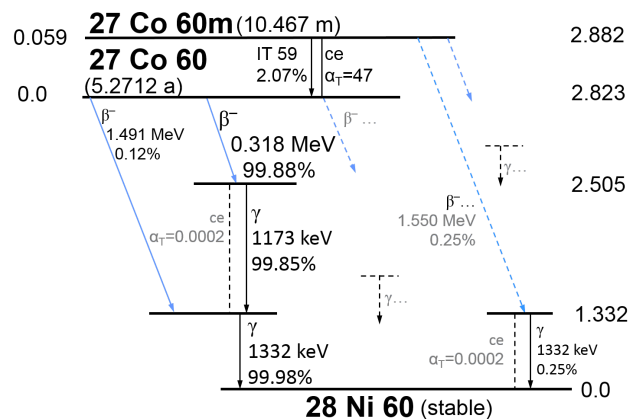
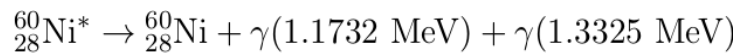


fig: Co^{60} decay energy levels

- I. ${}_{27}^{60}\text{Co}$ decays into an excited ${}_{28}^{60}\text{Ni}^*$ nucleus, producing a β and an antineutrino.



- II. Then the excited ${}_{28}^{60}\text{Ni}^*$ nucleus then decays into ground state ${}_{28}^{60}\text{Ni}$ nucleus in a 2 step process, producing two γ -rays of different energies.



4.1.2 Cesium-137 : It has a half life of ≈ 30.17 years . mainly undergoes 2 types of β -decay, releases 1 kind of γ -ray.

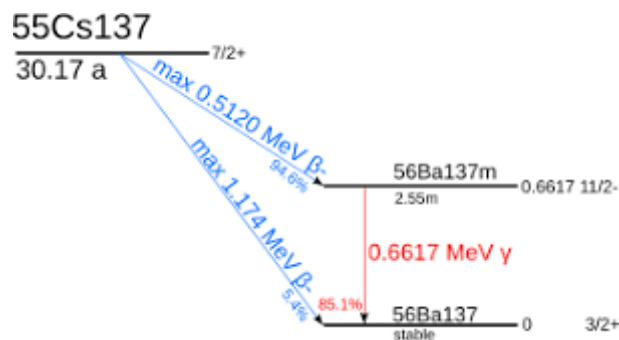
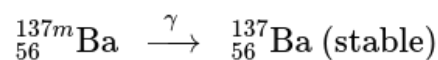
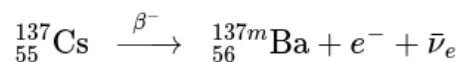


fig: Cs^{137} decay energy levels

- I. It emits a β particle (electron) with energy 0.512 MeV and gets to an unstable daughter ${}_{56}^{137m}\text{Ba}$ which later gets stable by emitting a γ -ray of 0.6617 MeV.



- II. It directly decays to stable ${}_{56}^{137}\text{Ba}$ (rare phenomena, around 5.4%).

4.2 Gamma interactions : High energy γ photons can interact with matter mainly in 3 ways.

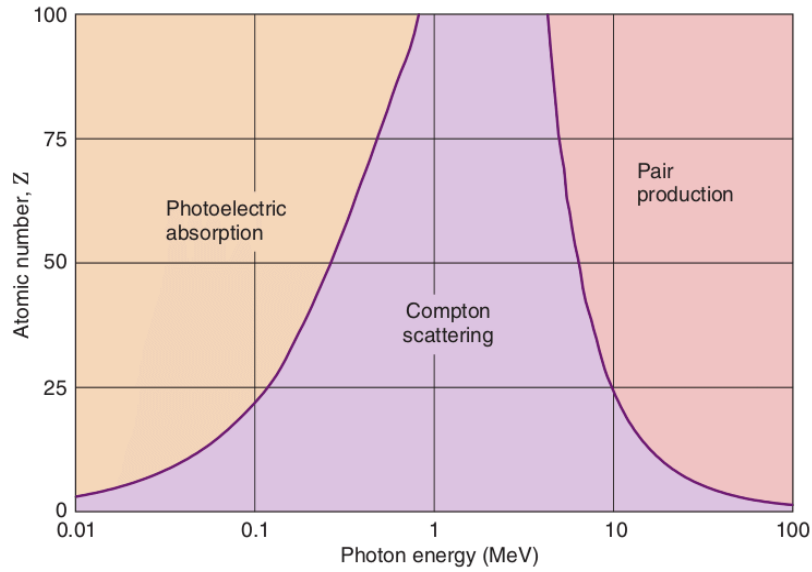


fig: Z – E Diagram of photon interactions

4.2.1 Photoelectric Effect :

Complete absorption of photon resulting ejection of an electron.

$$E_{kin} = h\nu - E_B$$

The process only takes place with K or L-electrons and occurs more often with substances with a high atomic number (Z). When the photon energy is too high, a nucleus with a high Z cannot handle the surplus of impulse either, that is why the photoelectric effect only occurs up to a limited energy value.

4.2.2 Compton Scattering :

Photon partially transfers its energy to electron, changing the photon's direction.

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{mc^2}(1 - \cos\theta)}$$

4.2.3 Pair Production :

Only if γ -ray energy is above 1.02 MeV, photon converts into electron and positron.

4.3 Scintillation counter : It emits flashes of visible light whenever a photon (here γ -ray) deposit energy on it. γ -ray can interact in all of the 3 ways mentioned earlier.

4.4 Photomultiplier Tube (PMT) : It takes visible light photons emitted by scintillator and converts them into electrons via photoelectric effect.

Then those electrons are multiplied through a series of dynodes which eventually results in a noticeable current pulse. The efficiency of a PMT tube is about 30%.

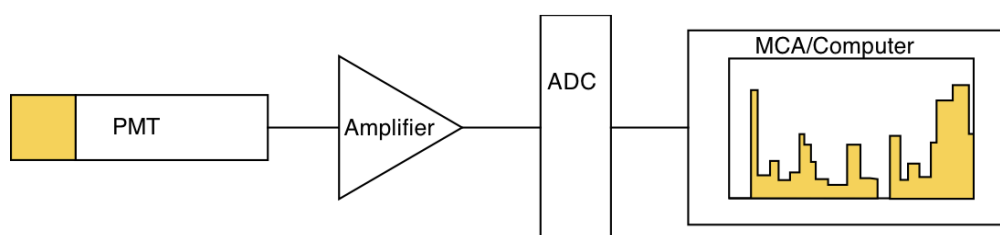


fig: Schematic of the data acquisition with PC

4.5 Multi Channel Analyzer (MCA) : The MCA counts number of counts for all the relevant energy ranges at the same time. As pointed out in the theory section. Energy of the detected signal is proportional to channel number. The proportionality constant is determined experimentally. Since we know two energy value E_A and E_B for Co^{60} we can find corresponding channel number A and B to find this proportionality constant α , say

$$\alpha = \frac{E_A - E_B}{A - B}$$

5 Data In Tabulated Form :

Here is the partial data table of Co^{60} spectrum:

Channel	Counts at 560V	Counts at 570V	Counts at 580V	Counts at 590V	Counts at 600V	Counts at 610V	Counts at 620V	Counts at 630V	Counts at 640V	Counts at 650V	Counts at 660V
0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0
4	1915	1726	1548	1467	1299	1201	1047	1019	1238	1550	2111
5	2169	1982	1847	1624	1551	1391	1379	1271	1882	2781	4222
6	2353	2139	1917	1692	1549	1427	1404	1410	1882	3055	4965
7	2564	2433	1987	1927	1787	1649	1511	1479	1502	2020	3094
8	2680	2447	2269	1959	1837	1679	1579	1449	1351	1521	1748

The Full data table (Along with Cs^{137} spectrum) can be found here:

https://docs.google.com/spreadsheets/d/1pVOExt2RtH3ocd-z-c6_DgYwf8RZqf9zCveB7YqRTtw/edit?usp=sharing

6 Analysis :

6.1 Graphs :

These are graphs of Channel Vs count with different HV of Co^{60} 's decay recorded for 10 minutes each.

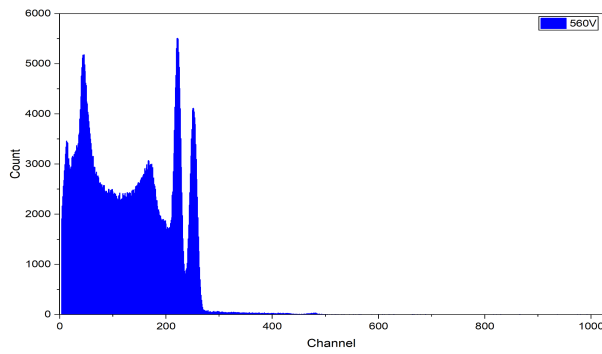


fig: Channel vs Count plot of Co^{60} at 560 V

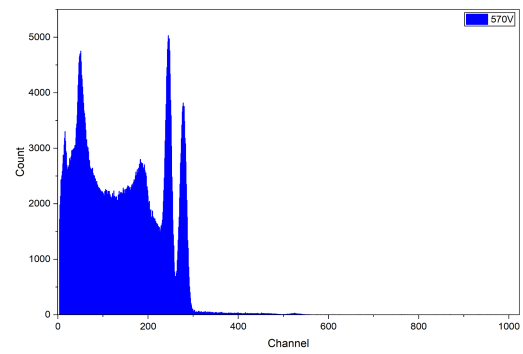


fig: Channel vs Count plot of Co^{60} at 570 V

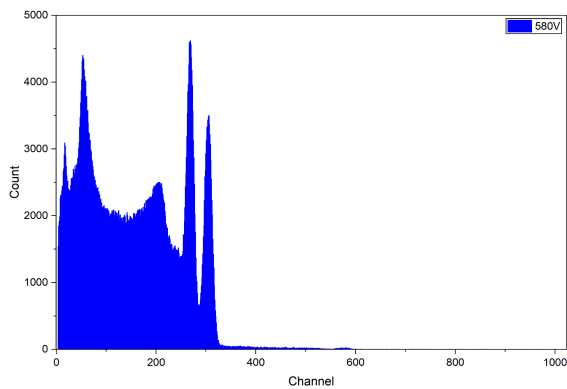


fig: Channel vs Count plot of Co^{60} at 580 V

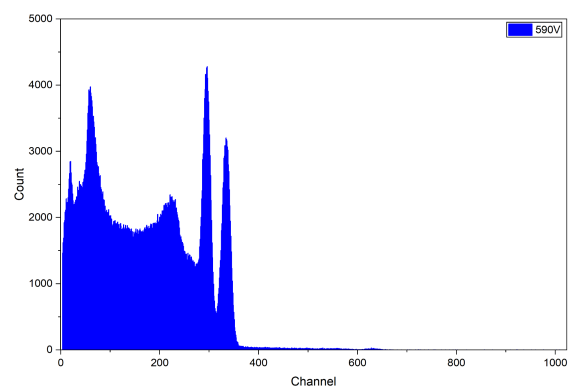


fig: Channel vs Count plot of Co^{60} at 590 V

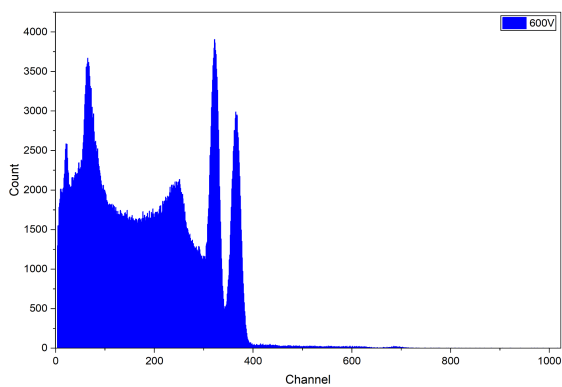


fig: Channel vs Count plot of Co^{60} at 600 V

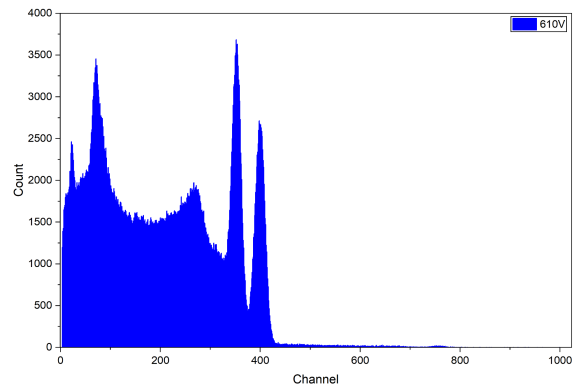


fig: Channel vs Count plot of Co^{60} at 610 V

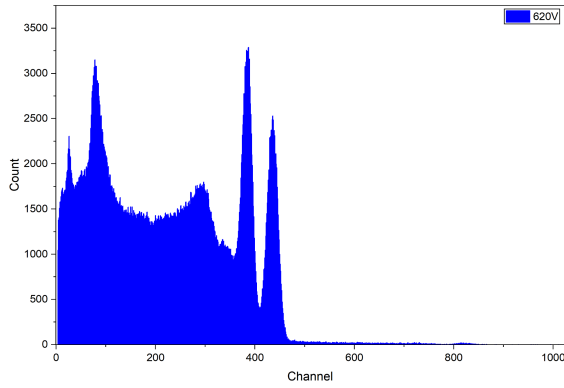


fig: Channel vs Count plot of Co^{60} at 620 V

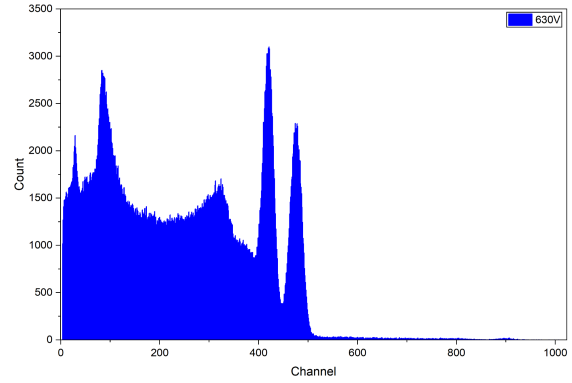


fig: Channel vs Count plot of Co^{60} at 630 V

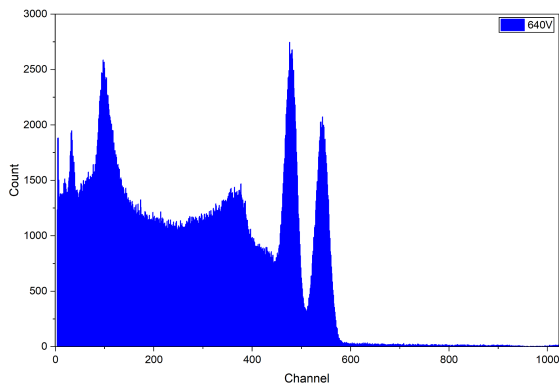


fig: Channel vs Count plot of Co^{60} at 640 V

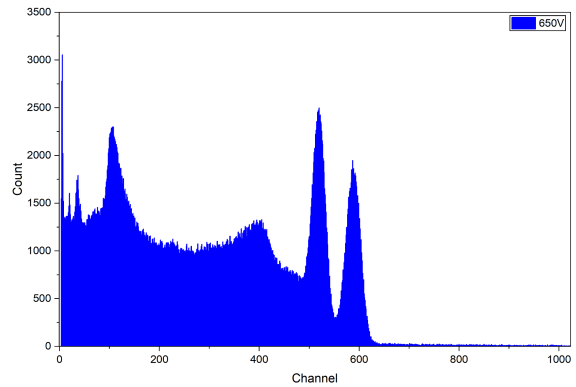


fig: Channel vs Count plot of Co^{60} at 650 V

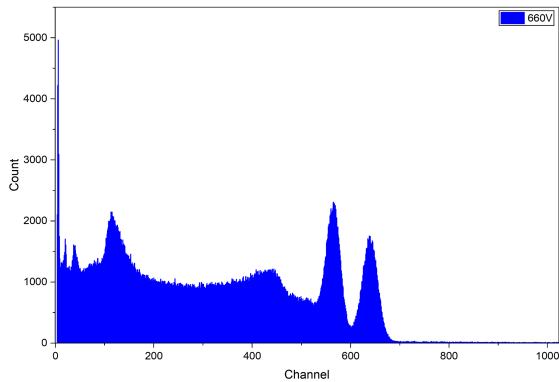


fig: Channel vs Count plot of Co^{60} at 660 V

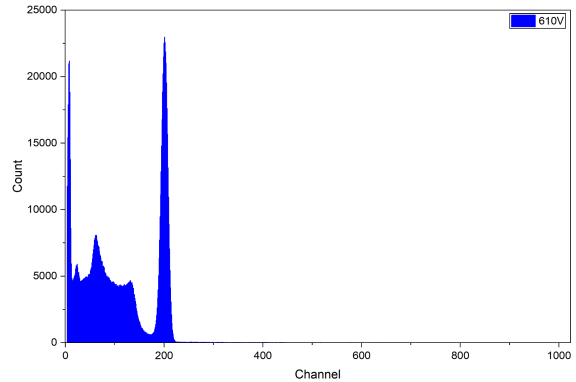


fig: Channel vs Count plot of Cs^{137} at 610 V

6.2 MCA Energy Calibration :

To calibrate the channels with energy we will use 610V data of Cobalt spectrum. Now as we know the value of E_A and E_B for Co^{60} is respectively 1.17 MeV and 1.33 MeV. From our data analysis software, we get the peaks:
Peak A: Channel 352 (corresponding 1.17 MeV)
Peak B: Channel 398 (corresponding 1.33 MeV)

Calibration constant :

$$\alpha = \frac{E_a - E_b}{A - B} = \frac{1.33 - 1.17}{398 - 352} = 0.00348 \text{ MeV/channel}$$

8.2.1 Photopeak energy of Cs^{137} :

Analyzing the Cs^{137} decay plot at 610V, we found the peak at channel 201.

So, photopeak energy:

$$E_{Cs} = \alpha \times 201 = (0.00348 \times 201) \text{ MeV} = 0.699 \text{ MeV}$$

6.3 Photopeak Efficiency of Co^{60} (at 610V) :

As we know,

$$\text{Photopeak Efficiency} = \frac{\text{Area Under The Peak}}{\text{Area Under The Whole Spectrum}}$$

To find the area under spectrum, we use integration, and obtain these following results

PMT Voltage (V)	Area Under The Curve			Photopeak Efficiency	
	Whole Spectrum	1.17 MeV Peak	1.33 MeV Peak	1.17 MeV Peak	1.33 MeV Peak
560	733918	5510	4115	0.00751	0.00561
570	735566	5032	3819	0.00684	0.00519
580	737700	4618	3480	0.00626	0.00472
590	740759	4280	3202	0.00578	0.00432
600	739252	3906	2990	0.00528	0.00404
610	744191	3683	2714	0.00495	0.00365
620	743780	3288	2529	0.00442	0.0034
630	744390	3100	2290	0.00416	0.00308
640	748906	2747	2047	0.00367	0.00273
650	751212	2498	1949	0.00333	0.00259
660	759612	2307	1757	0.00304	0.00231

6.4 Detector Resolution vs. PMT voltage :

If ΔE represents FWHM, and E represents photopeak channel number, We can find the resolution using:

$$R = \frac{\Delta E}{E} \times 100\%$$

Now , as we know the Cobalt spectrums had 2 peaks, so we fitted those data's baselines with parabolic function. Thus, by fitting with the sum of 2 independent gaussians, we obtain the channel number for photopeak and FWHM.

High Voltage	$\Delta E1$	E1	Resolution (R1)	$\Delta E2$	E2	Resolution (R2)
560	17.20465	5280	0.00326	15.30665	4170	0.00367
570	19.0496	4800	0.00397	16.58718	3820	0.00434
580	20.26969	4420	0.00459	18.18441	3520	0.00517
590	21.79945	4080	0.00534	19.89455	3220	0.00618
600	24.50513	3690	0.00664	21.44226	2970	0.00722
610	26.14058	3460	0.00756	23.60927	2700	0.00874
620	28.68317	3130	0.00916	25.62349	2480	0.01033
630	28.22747	2970	0.0095	29.60375	2260	0.0131
640	35.48782	2530	0.01403	31.3569	2010	0.0156
650	37.06132	2370	0.01564	34.24306	1860	0.01841
660	39.48679	2200	0.01795	36.95917	1700	0.02174

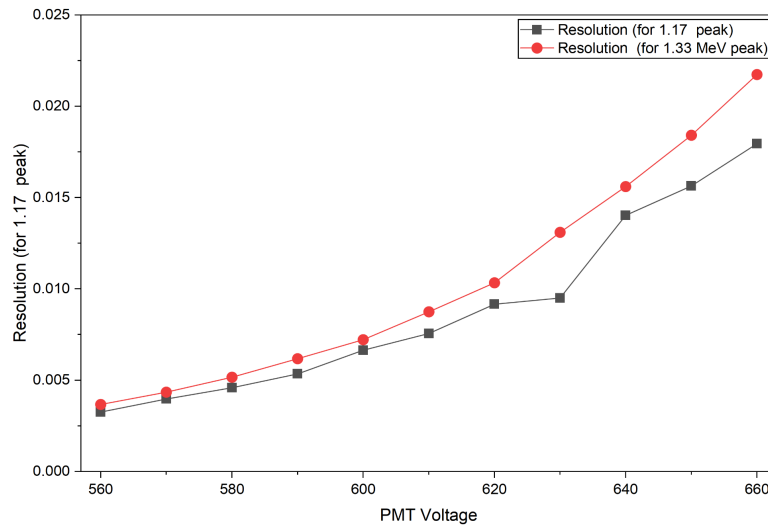


Fig: Plot of resolution of Co^{60} peaks at different PMT voltage

6.5 Analysis of Co^{64} spectrum :

For the Co-60 spectrum at 610V, the photopeaks for the 1.1732 MeV and 1.3325 MeV gamma rays were identified. Using the calibration constant $\alpha = 0.00348$ MeV/channel, the Compton edges are calculated to be at Channel 284 (0.9635 MeV) and Channel 330 (1.118 MeV). The backscatter energies for both gamma rays (0.2098 MeV and 0.2144 MeV) produce a combined

peak expected around Channels 62-63. Double escape peaks from pair production are predicted at Channel 45 (0.1512 MeV) and Channel 92 (0.3105 MeV). These calculated positions correspond well with the features observed in the experimental spectrum.

7 Conclusion:

This experiment successfully characterized the gamma-ray spectra of Co^{60} and Cs^{137} using a Multi-Channel Analyzer. The energy of the Cs^{137} Photopeak was experimentally determined to be 0.699 MeV, which deviates from the accepted literature value by 5.59%. An analysis of the detector resolution at various PMT voltages showed that 560V provided the optimal resolution within the tested range. The experiment also confirmed the expected behavior that increasing the PMT voltage increases system gain, causing the photopeaks to shift to higher channel numbers.